



Q. Which of the following is indication for non-invasive ventilation?

- a. Acute hypercapnic respiratory failure
- b. Acute exacerbation of bronchial asthma
- c. Facial burns
- d. Myocardial infarction with cardiogenic shock

Ans: (a)

Q. Identify the device shown below:

Ans. High-flow nasal cannula



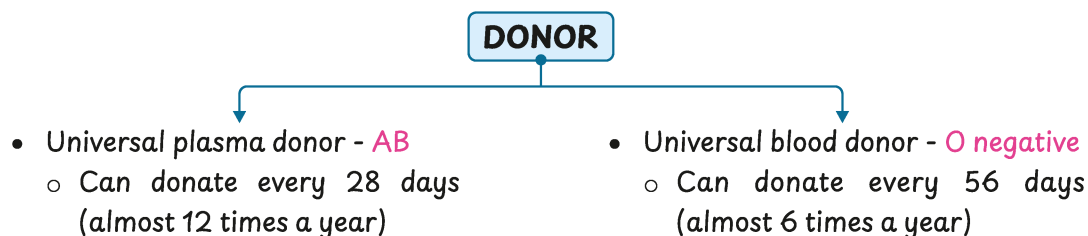


MASSIVE TRANSFUSION PROTOCOL

MASSIVE TRANSFUSION PROTOCOL

BASICS

00:00:41



MASSIVE TRANSFUSION

- >10 units of whole blood transfusion in 24 hours
Or
- Replacement of 50% of adult blood volume/ >2.5L of whole blood replaced in 4 hours
Or
- 6 units of fresh frozen plasma + 6 units of packed RBCs + 6 units of platelet-rich plasma to be given in ratio of **1:1:1** or **1:1:0.25** (If S.D.P. is used)
 - 4-6 Platelet-rich plasma (P.R.P.) = 1 Single Donor Plasma (S.D.P.)

6 units of packed red blood cells (RBCs)



6 units of fresh frozen plasma (FFP)



6 units of platelet condensed into 1 bag



- In patient with dengue hemorrhagic fever/DIC - give platelets
- In a bleeding patient, first give **packed RBC** because

Bleeding → Loss of blood → Circulatory failure → Metabolic Acidosis → Impairs ability of blood to clot

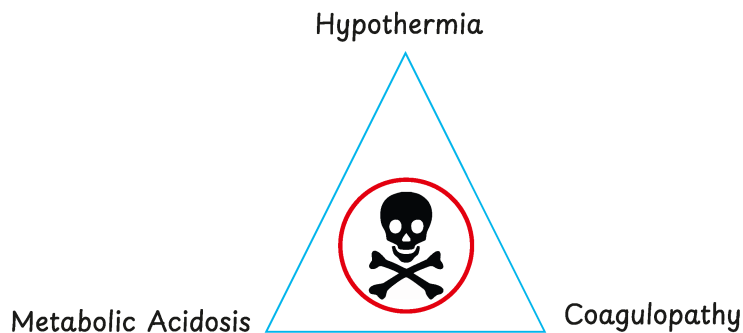
LIMITATIONS OF HUMAN CLOTTING SYSTEM

00:06:31





LETHAL TRIAD OF TRAUMA-INDUCED COAGULOPATHY



TRIGGERS FOR MASSIVE TRANSFUSION PROTOCOL

00:08:44

ASSESSMENT OF BLOOD COMPONENT SCORE (ABC SCORE)

Penetrating injury	1 Point
FAST +	1 Point
HR >100/min	1 Point
SBP <90 mmHg	1 Point

- Score >2 - decision making process for initiating transfusion

SET-UP

- Pressurized rapid transfusion (in-line warmer)
- Tranexamic acid - 1gm I/V
 - MOA - Anti-fibrinolytic agent



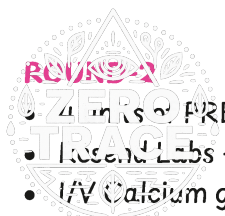
ADULT MASSIVE TRANSFUSION PROTOCOL

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- Trigger
 - ABC Score >2
 - Surgery/ GI bleeding/ OBS bleeding/ Trauma bleeding + Hypotension

ROUND-1

- Call blood bank, assign team members as A, B, C
 - A - Administer blood components
 - B - Record keeping and sampling
 - C - Runner
 - I/V Tranexamic acid - 1gm I/V stat
 - Connect pressurized transfuser
 - Give 4 units of packed RBC and 2 units of FFP
- If still no improvement, move to round-2



• 4 units of FFP, 4 units of FFP, and 1 unit SDP

- Resend Labs - Hematocrit, hemoglobin, metabolic profile, S. calcium (for citrate toxicity) & S. fibrinogen
- I/V Calcium gluconate (To prevent **laryngospasm/ tetany**)
- Cryoprecipitate if fibrinogen value <100 mg/L
- If still no improvement, move to round-3

ROUND 3

- Repeat round 2
- Give Factor VIIa for surgical hemostasis
- Ensure body temperature 37°C & pH = 7.35-7.45

HYPOVOLEMIA

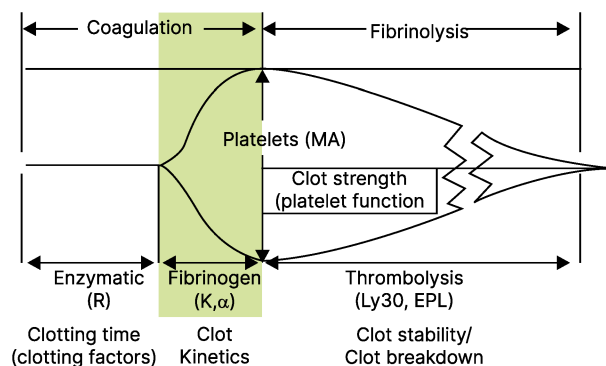
- All efforts made to stop bleeding rather than replace blood loss
- Blood loss of 150 ml/min → loss of **3L** blood volume in **20** minutes → B.P. becomes unrecordable → death by exsanguination

HYPOTHERMIA

- Erythrocytes stored at **1°C - 6°C**, packed RBCs, cold blood products → hypothermia
- Hypothermia → interferes with ability of clotting system to function optimally

HYPO-COAGULABILITY

- Special test - Thromboelastography
 - Use - To evaluate ability of blood to clot
- Serum fibrinogen levels checked
- Each **10-12** units of packed RBCs - a/w decrease in platelets
 - Causes transfusion associated thrombocytopenia
- Hypocoagulability ↑ chances of bleeding in patient



COMPLICATIONS OF MASSIVE BLOOD TRANSFUSION

00:17:12

- Coagulopathy
 - Due to ↓ platelets, ↓ fibrinogen & hypothermia
- TRALI
 - **2nd M/c** Cause of death **after blood transfusion**
 - More common with **FFP**
 - Occurs **<6 hours** of transfusion
- TACO - **M/c** Cause of death after blood transfusion
- Hyperkalemia - Diastolic arrest
- Hypocalcemia, Tetany - Citrate toxicity
- Both TRALI & TACO contribute to respiratory distress in patient



	BP	JVP	BNP
	Normal	Normal	Normal/↑
	↑	↑	↑↑

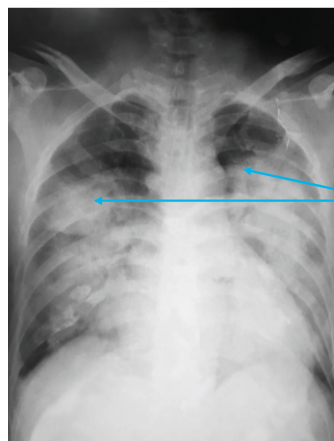
MCQs

Q. In a patient of burns, after 2 hours of receiving FFP, sudden onset of distress, hypoxemia, tachypnea, hypotension, and B/L pulmonary infiltrates on chest X-ray. What is the most likely diagnosis?

- a. TRALI
- b. TACO
- c. Tetany
- d. RDS

Ans. (a)

- Also known as Shock Lung
- Example of non-cardiogenic pulmonary edema

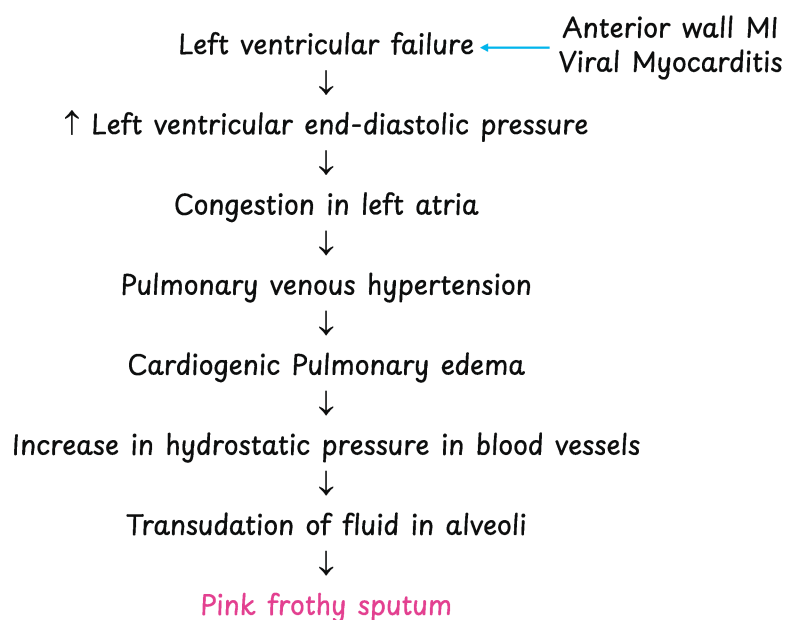


B/L Infiltrates

Chest X-ray

CARDIOGENIC PULMONARY EDEMA

00:00:30



- **Gross elevation of BNP levels:** Helps to differentiate cardiogenic pulmonary edema from non-cardiogenic pulmonary edema

ARDS/SHOCK LUNG

00:03:06

- Mortality rate: **46%**
- Leading indirect cause of ARDS : **Sepsis**
- Leading direct cause of ARDS : **Pneumonia**
- Direct causes are more common than indirect cause
- 10% of all ICU admissions are due to ARDS

SEPSIS 3 GUIDELINES

- **Older definition:** ≥ 2 -point increment in the Sequential Organ Failure Assessment (SOFA) score in documented/suspected infection
- Organs involved in the SOFA score:

ORGAN	PARAMETER	SCORE
Brain	Glasgow Coma Scale (GCS)	0-4
Heart	Mean Arterial Pressure (MAP)	0-4
Lungs	PaO ₂ /FiO ₂ ratio	0-4
Liver	Serum Bilirubin	0-4
Kidney	Serum Creatinine & Urine Output	0-4
Hematological parameters	Platelet Count	0-4

- Total Score Range: 0 to 24

SYSTEMIC INFLAMMATORY RESPONSE SYNDROME

- Components:
 - Fever : $>38^{\circ}\text{C}$ or $<36^{\circ}\text{C}$
 - Hypothermia - fall in core temperature $<35^{\circ}\text{C}$ in an adult
 - Heart rate $>90/\text{min}$ (d/t hyperdynamic circulation)
 - Respiratory rate $>20/\text{min}$
 - Total leukocyte count : **Leucocytosis** ($>12,000/\text{cu.mm}$) or **leukopenia**
- **Older definition of sepsis:** Sepsis = $\geq 2/4$ parameter (above mentioned) present + Documented/Suspected infection

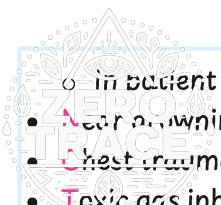
SEPTIC SHOCK

- Definition: Documented infection along with:
 - Requirement of **vasopressor** support to maintain mean arterial pressure >65 mmHg
 - Serum lactate ≥ 2 mmol/L
- T/t:
 - 1st line management: Normal saline (**preferred**)
 - Preferred vasopressor: Norepinephrine ($\uparrow$ diastolic BP & coronary blood flow)
 - No role of colloids (Dextran, Albumin, HES)

CAUSES OF ARDS

00:13:39

DIRECT CAUSES	INDIRECT CAUSES
Mnemonic - PAN-CT • Pneumonia - M/C/C of ARDS • Aspiration of gastric contents into lung → Chemical pneumonitis (Mendelson syndrome)	Mnemonic - STD-P • Sepsis • Trauma ◦ Multiple bone fracture



- In patient with full stomach trying to be intubated
- Near drowning
- Chest trauma → Pulmonary Contusion
- Toxic gas inhalation
 - Bhopal gas tragedy – Methyl isocyanide gas
 - Chernobyl accident
- Flail chest
- Head trauma
- Burns
- **Transfusion-related acute lung injury (TRALI)**
 - Seen with FFP (Fresh Frozen Plasma) > Blood transfusion
- Drug overdose
- Acute Pancreatitis
- Post-cardiopulmonary bypass

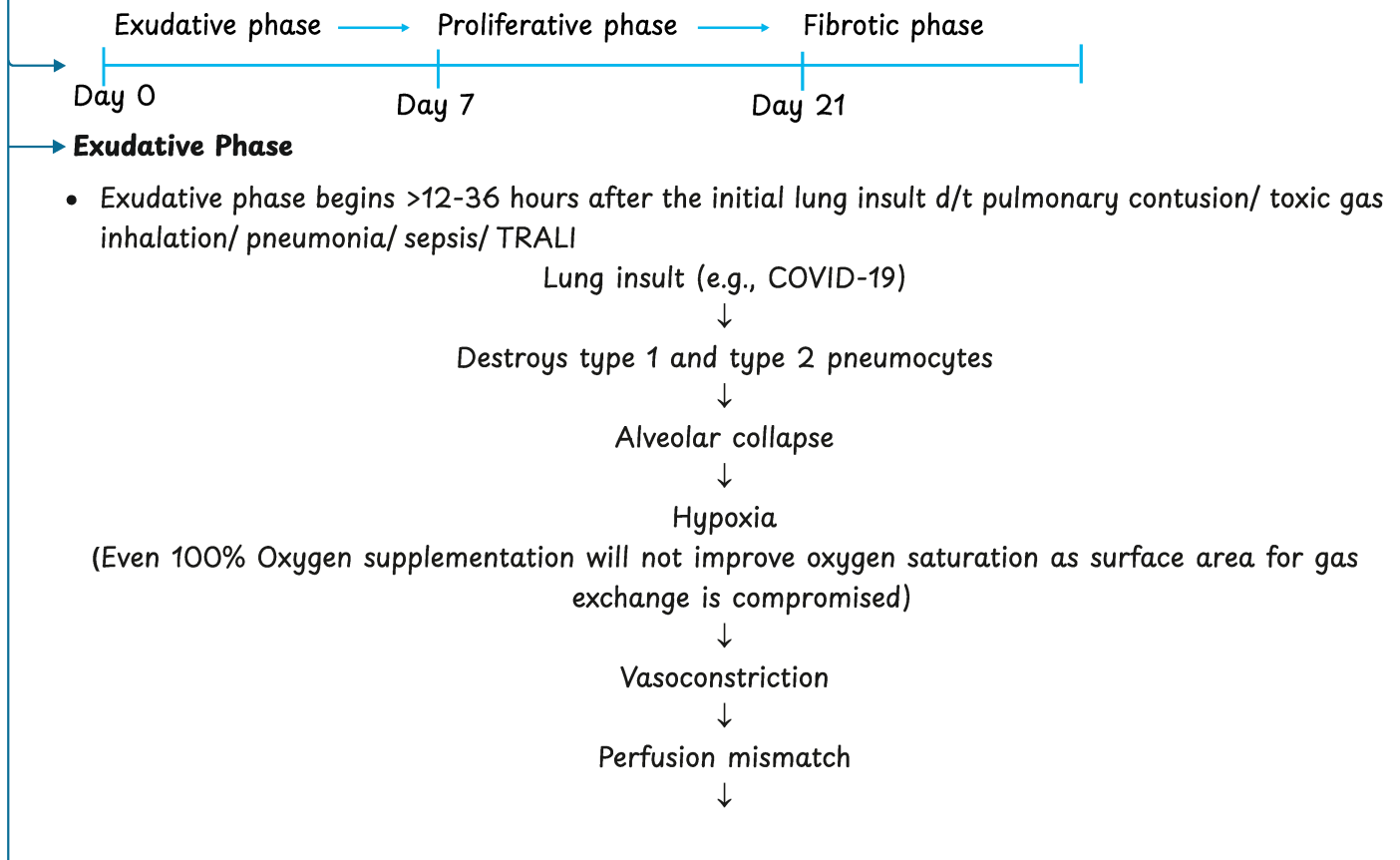
- Hydrocution: If a person dives into cold water with face down/on his belly → Activation of vagus nerve → Cardiac arrest

Important Information

- **Fat Embolization Syndrome (FES):** Diagnostic Criteria: Gurd's Criteria
 - Require 1 major & 4 minor criteria for diagnosis of fat embolism
 - Some Key Features: CNS malfunction like coma, Petechial developing in axilla, involving chest wall
- Respiratory Distress Within 6 Hours Post-Transfusion is mainly caused by:
 - TRALI (Transfusion-Related Acute Lung Injury)
 - TACO (Transfusion-Associated Circulatory Overload): **Leading cause of death after blood transfusion**

PATHOPHYSIOLOGY OF ARDS

00:25:15





Pulmonary microthrombi formation (add to dead space)



Endothelial cells (most susceptible to hypoxia)
become leaky, and fluid starts accumulating in the alveoli



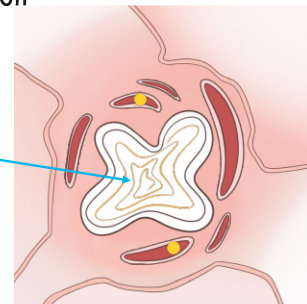
Non-cardiogenic pulmonary edema (high protein)



Diffuse alveolar damage (Thin, slimy membrane)



Further hampers gas exchange
(Patient fails to respond despite best ventilator support)



→ **RADIO-PATHOLOGICAL CORRELATES OF EXUDATIVE PHASE**

- Collapsed alveoli lined by whorls of hyaline membrane
- Pulmonary microthrombi formation
- High-protein pulmonary edema
- Intra-pulmonary shunting (No improve in saturation even with 100% Oxygen supplementation)
- Reduced lung compliance & increased work of breathing
- Radiological findings that develop due to these changes are:
 - CXR shows **B/L infiltrates**
 - HRCT shows **dependent pulmonary edema**
→ O₂ saturation improves with prone position

Important Information

- Invention highly beneficial for RDS
 - Low volume ventilation
 - Prone position

→ **Proliferative Phase**

- Type 2 pneumocytes start regenerating/proliferating
 - ↓
 - Surfactant production resumes
 - Some type 2 pneumocytes differentiate into type 1 pneumocytes
 - ↓
 - Gas exchange surface area regenerates
- } Chances of recovery

→ **Fibrotic Phase**

- Lungs will become smaller d/t fibrosis
- Exercise capacity declines
- Bulla formation may occur d/t alveolar duct damage

DIAGNOSTIC CRITERIA

00:36:25

- $P(a)_{O_2}/F_{iO_2}$ ratio: ↓ **within 7 days** of the initial lung insult
- **Normal:** $100 \text{ mmHg} / 0.2 = 500 \text{ mmHg}$

Mild ARDS	<300 mmHg
Moderate ARDS	100-200mmHg
Severe ARDS	< 100mmHg

- Chest X-ray or CT scan: B/L opacities, pulmonary edema not explained by any cardiogenic involvement (not explained by volume overloading)
- **Echocardiography:** Normal left atrial pressure or absence of left atrial hypertension



Chest X-ray

Important Information

- TACO (Transfusion associated circulatory overload) → Cardiogenic pulmonary edema

TREATMENT OF ARDS

00:41:37

- General supportive measures:
 - Prevention of deep vein thrombosis
 - Enoxaparin OD
 - Pneumatic compression stockings (powered by air pumps)-compress calf muscles & ↓ chances of stasis & further pulmonary embolism minimized
 - Prevention of gut bleeding: PPIs
 - Prevention of Central Line-associated Bloodstream Infection (CLABSI) -associated with *S. aureus*
- **Low volume ventilation** : Mortality reducing intervention
 - Tidal volume: 6ml/kg, lean body weight
 - To prevent barotrauma to damaged alveoli
 - Normal TV for routine ventilation: 12ml/kg
- E.g., in GBS d/t diaphragmatic paralysis
 - Plateau pressure: <30 cm H₂O (to prevent tension pneumothorax)
 - Respiratory rate: <35/min
 - FiO₂: ~ 0.6
 - Target SpO₂: 88-95%
 - High PEEP: ≥5cm H₂O
- Fluid management: Restricted IV fluids
- Prone position: Improve dependent pulmonary edema
- Extracorporeal membrane oxygenation (ECMO)
- Steroids: No role in non-COVID-19 causes of ARDS
 - Drugs used for management of COVID-19 - Methylprednisolone + enoxaparin
- M/C/C death in ARDS : **Sepsis** >non-pulmonary causes like advanced age



Q. All of the following are features of ARDS, except-

- a. High protein pulmonary edema
- b. Low pO_2 / FiO_2 ratio
- c. Elevated left atrial pressure
- d. Bilateral pulmonary opacities

Ans. (c)



3R. SNAKE BITE



SNAKE BITE

00:00:40

- It is a neglected tropical disorder
- 300 species are present in India out of which 60 odd species are venomous

INCIDENCE

- Number of deaths in India: >50,000/year
- Population exposed to snake bite: >50 million
- Age: 20-55 years
- Men are at higher risk because of occupational exposure in agriculture
- Non-venomous snake bites are more common (70%)
- If a person is wearing shoes, then the fangs of the snake may not be able to penetrate the skin or administer venom, or if the angle of the bite is not adequate; such bites are known as **dry bites**
- Overall, 85% of all snakebites are harmless

Important Information

Indian rat snake (*Ptyas mucosa*)

- Non-venomous snake
- 6-8 feet long
- Doesn't form a hood
- Lower lip has black bands
- Diurnal habits



THE BIG FOUR



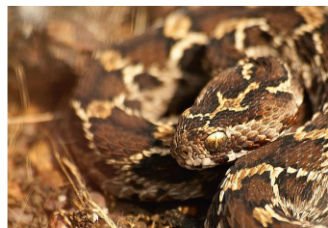
Cobra



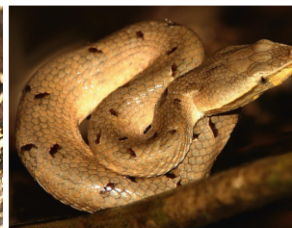
Krait



Russell's Viper



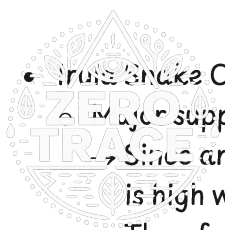
Saw Scaled Viper



Humped Nose Viper

- Cobra – Neurotoxic venom
- Krait – Neurotoxic venom
- Russell viper – Hemotoxic > Neurotoxic venom
- Saw scaled viper – Hemotoxic
- Bite to needle time: **<4 hr**
- Anti-snake venom does not work against **Humped nose viper** (Hemotoxic venom)

Free By - ZeroTrace



- True Snake Catchers Industrial Cooperative Society, Chennai
 - Major supplier of venom for both research & for inoculation into equines for raising antibodies
 - Since antibodies are produced in equines, the chances of allergic reactions (30-40%) and anaphylaxis is high when it is administered to humans
 - Therefore, Adrenaline should always be available when ASV is given
- ASV is made in a lyophilized form
 - Shelf life: 5 years
 - Stored at: $<25^{\circ}\text{C}$
- Liquid ASV
 - Stored at: $<2-8^{\circ}\text{C}$

FIRST AID FOR SNAKE BITE

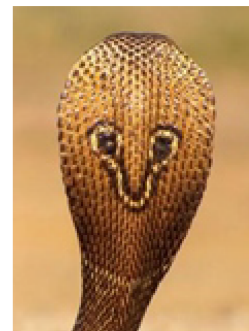
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- **R** - Reassurance the patient
- **I** - Immobilize the limb same way as a fractured limb
 - Do not walk
 - Use bandage/cloth to hold splints, but do not block the blood supply due to risk of ischemia
 - Do not apply any tight ligature or compression bandage, as that can increase risk of digital gangrene
- **GH** - Get to the hospital immediately
- **T** - Tell the doctor of any systemic symptoms that manifest on the way to the hospital
- No tourniquet (Increases risk of digital gangrene)
- No cutting
- No electrocautery
- No walking (Increases the risk of spread of venom & progression of paralysis will be faster)
- Always position the B.P. cuff proximally before cutting the tourniquet
 - Abrupt removal of the tourniquet may result in a sudden influx of venom into the bloodstream
 - It may consequently lead to respiratory paralysis/arrest
 - Venom triggers histamine release, leading to a sudden drop in BP
- Nitrate spray: Efficacy +/-
- Mainstay of treatment: Anti-Snake Venom - approx. 20 vials (India)

MECHANISM OF ACTION OF SNAKE VENOM

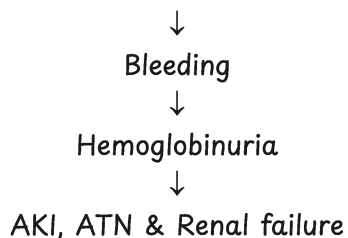
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- Cobra bite:
 - Cobra can be identified by spectacle/binocellate mark on the hood
 - **Post-synaptic toxin**: More Ach release
- Krait bite
 - **Pre-synaptic toxin**: Less Ach release
- Russell viper
 - Neurotoxicity: Presence of Phospholipase- A_2
 - Damages cell membrane of nerve cells
 - Synaptic vesicles will not be able to fuse to Ach release
 - Hemotoxicity: Activation of factor X



↓
Activate coagulation cascade
↓
Consumption of coagulation factors

Free By - ZeroTrace

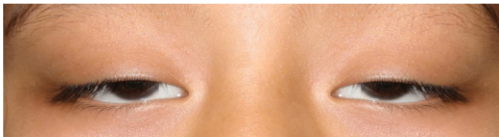


VIPER ENVENOMATION MANIFESTATIONS

- Extreme local pain
- Regional tender lymphadenopathy

BLEEDING MANIFESTATIONS	• Subconjunctival hemorrhage • Purpura • Epistaxis • Gum bleeding • GI bleeding • Retroperitoneal bleeding
 Bleeding Gums	
GIT MANIFESTATIONS	• Vomiting • Excruciating abdominal pain • Retroperitoneal bleeding • Fall in B.P
KIDNEY MANIFESTATIONS	• Acute Kidney Injury • Acute Tubular Necrosis

ELAPID EVENOMATION MANIFESTATIONS

4P's	4D's
Ptosis  Ptosis	Diplopia
Paralysis of jaw & tongue	Dysarthria/Anarthria
Pooling of secretions (Frothing at corners of mouth & rattling sound from throat, bilaterally conducted sounds on auscultation)	Dysphagia
Paradoxical breathing	Dyspnea

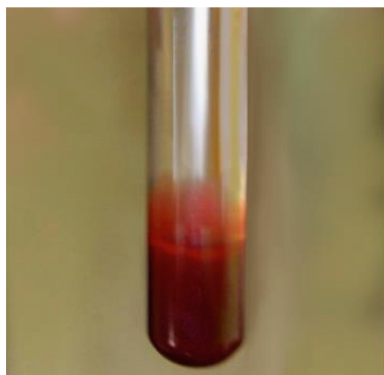


Whole Blood clotting test (WBCT)

20 MIN WHOLE BLOOD CLOTTING TEST (20 WBCT)

Collection

Draw 2 ml of venous blood and transfer directly into a lean and dry glass tube. Leave it upright, open, undisturbed for 20 and/or 30 minutes at room temperature



Normal

After exactly 20 minutes, pick up the tube and invert it. If a solid clot is retained, the test indicates normal coagulation. (Grade 0, no coagulopathy)



Abnormal

If the clot breaks down quickly upon inversion of the tube or fails to coagulate, the test indicates a coagulopathy



- CBC
- LFT
- KFT
- Coagulogram
 - Bleeding time: Normal – 2 to 9 minutes
 - Prothrombin time: Normal – 11 to 16 seconds
 - Activated partial thromboplastin time: Normal – 30 to 45 seconds

MANAGEMENT

00:21:33

- Pain: Paracetamol or Tramadol (for severe pain)
- Apply B.P. cuff proximal to the tourniquet to prevent a gush of venom into the circulation
- **Polyvalent Anti-Snake Venom (10-30 vials)**
 - The dosage for children and pregnant individuals is equivalent to that administered to adults.
 - $T_{1/2}$ = 90 minutes
 - 1 mg vial of ASV neutralizes **6 mg of Russell viper venom**
 - Average number of vials used: 20 vials
- Primary allergic reaction to ASV:
 - Hydrocortisone 100 mg
 - H_1 blocker: Chlorpheniramine maleate (Avil)
 - **Adrenaline**
 - 0.5 mg IM undiluted (1:1000)
- Atropine + Neostigmine: Cobra bite

INDICATIONS OF ASV

• Glycopenia

• Neurotoxicity:

- Ask the patient to raise the neck
- Single breath count
- CVS abnormalities: Low B.P & arrhythmia (V.T & V.fib)
- Severe abdominal pain & vomiting (indicates retroperitoneal bleed)
- Local swelling >1/2 of circumference of arm or leg
- Rapid extension of swelling above the level of waist

CLINICAL FEATURES OF SNAKE BITE

00:27:44

FEATURES	COBRA	KRAITS	RUSSELL VIPER	SAW SCALED VIPER	HUMPED NOSE VIPER
Local Pain/Tissue damage	Yes	No	Yes	Yes	Yes
Ptosis	Yes	Yes	Yes	No	No
Hemostatic abnormality	No	No	Yes	Yes	Yes
Renal complication	No	No	Yes	No	Yes
Response to neostigmine	Yes	+/-	No	No	No
Response to ASV	Yes	Yes	Yes	Yes	No

RECOVERY PHASE

00:31:22

- If an adequate dose of appropriate ASV has been administered, the following responses may be seen:
 - Spontaneous systemic bleeding stops within 15-30 minutes
 - Neurotoxic envenomation improves in 30 minutes
 - Single breath count
 - Raise neck
 - Hypotension improves within 30 minutes
 - 20 WBCT recovery takes 6 hours



Q. What is the bite-to-needle time for administration of polyvalent snake venom in case of a snake bite?

- a. 4 hours
- b. 2 hours
- c. 6 hours
- d. 5 hours

Ans (a)

Q. Anti-snake venom is not effective against which of the following snake bites?

- a. Cobra
- b. Krait
- c. Russell viper
- d. Humped nose viper

Ans (d)

Q. All of the following are used to manage a case of snake bite, except?

- a. Immobilize the patient
- b. ASV within 4 hrs of snake bite
- c. Apply a tourniquet
- d. Check for clot features in the test tube after 20 minutes

Ans (c)



ACUTE HIGH ALTITUDE PULMONARY EDEMA AND HYPOTHERMIA



PATHOPHYSIOLOGY

00:00:17

Q. What is the type of hypoxia seen at high altitude?

- a. Hypoxic
- b. Hypobaric
- c. Anemic
- d. Histotoxic

Ans. (b)

- Atmosphere becomes thinned out as we go higher up

HYPOXIC HYPOXIA	Structural lung diseases contributing to hypoxia like Interstitial lung diseases, pulmonary fibrosis, silicosis, sarcoidosis, COPD
ANEMIC HYPOXIA	Patients with methemoglobinemia, carbon monoxide poisoning (majorly carboxyhemoglobin in body and less oxyhemoglobin)
HISTOTOXIC HYPOXIA	Cyanide poisoning affects functioning of cytochrome oxidase in mitochondria

- Inspiratory pressure at high altitude (Mount Everest)-43 mmHg

ACUTE MOUNTAIN SICKNESS

- Headache
 - Hypobaric hypoxia → cerebral vasodilatation → stretching of pain-sensitive structures in brain - Dura mater (innervated by trigeminal nerve)
- Shortness of breath
 - Hypoxia causing ventilation-perfusion mismatch
 - Exercise induced pulmonary vasoconstriction → pressure on pulmonary artery ↑ → pulmonary artery hypertension → capillary stress failure → leaky capillaries → pulmonary edema
 - Endothelial cells are most vulnerable to hypoxia → NO ↓ → hydrostatic pressure ↑
- Cheyne-Stokes breathing
 - Present in ↑ Intracranial pressure, congestive heart failure, high altitude pulmonary edema
 - Respiratory center not able to adapt to sudden low oxygen tension
 - Cheyne-Stokes breathing: Hypoventilation → Apnea → Hyperventilation
→ Hypoventilation leads to drop in SpO₂
- Disturbed sleep, fatigue, flatulence and flatus

Capillary Stress failure



PREVENTION OF ACUTE MOUNTAIN SICKNESS

00:10:19

- **Acetazolamide**

• MOA: **Causes bicarbonaturia**

- Inducing metabolic acidosis → protons will stimulate respiratory center → hyperventilation
- Alternate drug in case of Sulfa drug allergy: **Dexamethasone**

MILD ACUTE MOUNTAIN SICKNESS	MODERATE ACUTE MOUNTAIN SICKNESS (HAD O ₂)
• Discontinue ascent and immediate descent to below the level where the symptoms were first reported • Drug used: Acetazolamide (both for prevention and treatment of mild acute mountain sickness)	• Immediate descent for worsening symptoms • If descent is not possible → Gamow bag used → simulated descent upto 2500 m → Hyperbaric therapy • Acetazolamide and/or Dexamethasone • Low flow oxygen @2-4 L/min • Oxygen concentrators can be used to give oxygen to the patient



Gamow bag

Important Information

- Hyperbaric oxygen therapy used in gas gangrene

TREATMENT FOR HIGH ALTITUDE CEREBRAL EDEMA AND PULMONARY EDEMA

HIGH ALTITUDE CEREBRAL EDEMA	HIGH ALTITUDE PULMONARY EDEMA
• Vasogenic cerebral edema (interstitial edema of brain) • Immediate descent/evacuation ↓ - If not available ↓ - Patient placed in Gamow bag ↓ - Simulated descent • Oxygen: 2-4 L/min • DOC: Dexamethasone IV	• Immediate descent/evacuation ↓ - If not available ↓ - Patient placed in Gamow bag ↓ - Simulated descent • Oxygen: 4-6 L/min • DOC-Nifedipine: ○ To minimize/counterbalance capillary stress failure ○ Nifedipine is a calcium channel blocker-8 mg used ○ Also effective for prevention • Tadalafil and Sildenafil (Phosphodiesterase 5 inhibitors) can also be used ○ Increase the level of NO at level of endothelial cells

OTHER APPROACHES

00:20:46

- **Salmeterol** (LABA): Long-acting Beta 2 agonist
- Clearance of salt/water from alveoli → wetness of alveoli ↓ → ↓ work of breathing → ↓ stress on the lungs
- PDE-5 inhibitors: **Tadalafil**
- **Nifedipine** for prevention of HAPE in those who have to ascend rapidly

CLINICAL FEATURES

00:21:41

SYMPTOMS

- Young male/female, on day 2-4 of arrival at **2000-2500 m** starts having:
 - Fatigue, shortness of breath, ↓ exercise intolerance
 - Nonproductive cough (debilitating enough to cause a rib fracture)
 - Peri-oral cyanosis
 - Bloating/flatulence/flatus passing is ↑
- Physical examination and workup findings
 - B/L crackles, ↑ respiratory rate and ↑ heart rate
 - Altered mentation due to high-altitude cerebral edema
 - ABG- Hypoxia (pO₂ ↓) and pCO₂ ↓ → **Type I respiratory failure/ respiratory alkalosis**
 - Echocardiography- **Right ventricular strain** due to Pulmonary artery hypertension
→ Right ventricular failure due to non-cardiogenic cause

CHRONIC MOUNTAIN SICKNESS

00:24:54

- Also known as **Monge's disease**
- Seen in Tibet and Andes region
- **Erythrocytosis** due to chronic hypoxia
- Sluggish lung circulation and PAH
- Cor Pulmonale: RVF due to **non-cardiogenic** cause

TREATMENT

- Venesection
- Acetazolamide contributes to bicarbonaturia → metabolic acidosis induced stimulation of respiratory center → **pre botzinger complex** is stimulated → hyperventilation to overcome hypoxia

HYPOTHERMIA

00:27:31

- Defined as core temperature **<35°C**

HYPOTHERMIA	
PRIMARY ACCIDENTAL HYPOTHERMIA	SECONDARY
Worsened with: • Alcohol which causes cutaneous vasodilation → ↑ loss of heat from the body • Antidepressant • Antipsychotics	Systemic disorders: Myxedema coma